Learning Hodgkin-Huxley action potential model [1]

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[1] Following the origingal paper

Description of neural tissue

Neural tissue consists of *interconnected* specialized cells called neurons. A neuron has special structures: axon, and dendrites serving a unique purpose of transmitting *brain signals*.

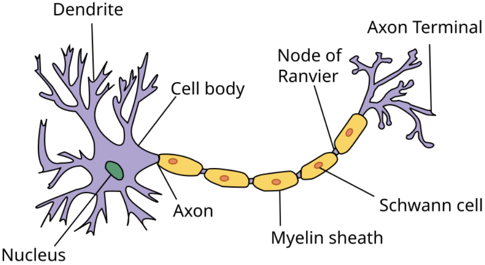

Figure: Structural parts of a neuron

Neurons are electrically excitable and can generate electrochemical pulse called an *action potential*. This potential travels rapidly along the axon and activates synaptic connections as it reaches them. The signal can propagate throughout the whole neural network.

Diverse system of brain signal propagation models

1. Single-Neuron Signal Propagation Models
   Focus on how electrical impulses (action potentials, AP) travel along individual neurons.
2. Multi-Compartment Neuron Models
   Account for neuron's structure and capture how signals attenuate and integrate across dendrites and axons.
3. Synaptic Transmission Models
   Describe signals propagation between neurons with electrical/chemical synapse models.
4. Network-Level Propagation Models
   Models average activity across populations or whole-brain networks (Wilson–Cowan, Jansen–Rit, Amari).

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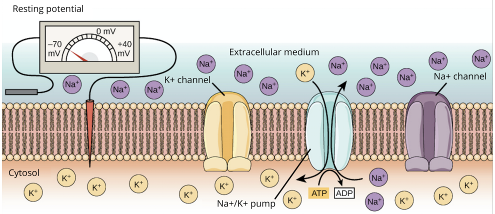
Action potential mechanism

Cell's membrane can be seen as a *condenser*. External plate has a greater charge compared to interior (Na^+ ions outside cell prevail over internal K^+ ions). Potential of the resting cell $v \approx -70$ mV.



At rest, all Na+ channels whose opening depends on the voltage and most of the K+ channels whose opening depends on the voltage are closed. Na+/K+ pumps transport K+ ions inside the cell and Na+ ions outside the cell.

Figure: Cell's membrane structure and environment

The membrane has voltage-gated ion channels (special proteins). If the membrane potential changes to a threshold level the channels open. Due to the concentration gradient, positive Na^+ ions rush inside the cell producing electrical current and further increasing v.

Action potential mechanism

Electrical current produce elector-magnetic field that increase potential further along the membrane. The signal propagates through the axon.

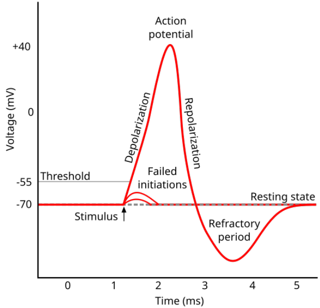


Figure: Action potential phases

After excitation, the neuron enters a state of refractoriness, when no signals can excite it again Then it enters a phase of relative refractoriness, when it can be excited only by extremely strong signals.

Membrane as an electrical circuit

The *Hodgkin–Huxley* model treats each component of an excitable cell as an electrical element:

1. The lipid bilayer is represented as a capacitance C_m.
2. Voltage-gated ion channels are represented by electrical conductances $g_n(t, v)$.
3. The electrochemical gradients driving the flow of ions are represented by voltage sources E_n whose voltages are determined by the ratio of the intra- and extracellular concentrations of the ions
4. Ion pumps are represented by current sources I_p

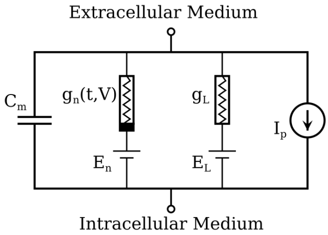


Figure: Hodgkin–Huxley model

Current equation

Kirchhoff equation for the circuit:

$$i(t) = C\frac{du(t)}{dt} + g_{Na}(t)(v(t) - E_{Na}) + g_K(t)(v(t) - E_K) + g_L(v(t) - E_L), \tag{1}$$

which can be rewritten as

$$\begin{aligned} i(t) &= C\frac{dv(t)}{dt} + g(t)(v(t) - E(t)), \text{ where} \\ g(t) &= g_{Na}(t) + g_K(t) + g_L \\ E(t) &= \frac{E_{Na}g_{Na}(t) + E_K g_K(t) + E_L g_L}{g_{Na}(t) + g_K(t) + g_L} \end{aligned} \tag{2}$$

Channel equations

Both Na^+ and K^+ channels are made up of 4 internal gates. All gates are open with some probability and are mutually independent. A channel is open only if all its gates are open. K^+ has four identical gates with opening probabilty n. Na^+ has 3 gates with the probability m and one with the h.

Dynamics of the opening probability $s(t), s \in \{n, m, h\}$ responds to the following equation:

$$\frac{ds(t)}{dt} = \alpha_s(v(t))(1 - s(t)) - \beta_s(v(t))s(t). \tag{3}$$

The $\alpha_s(v)$ and $\beta_s(v)$ are the opening and closing rates that depend solely on the voltage applied to the membrane. The total conductance of each channel is proportional to the average number of open channels:

$$g_{Na}(t) = \bar{g}_{Na} m^3(t) h(t), \tag{4}$$

$$g_K(t) = \bar{g}_K n^4(t), \tag{5}$$

where $\bar{g}_{Na}$ and $\bar{g}_K$ are the maximum conductance of the channel.

Learning opening/closing rates

It is possible to estimate each $\alpha_s(v)$, $\beta_s(v)$ in a careful experiment dissecting each channel individually. Instead, we can parameterize these functions and learn them altogether from measurements of the $v(t)$ and $i(t)$ on the whole membrane. The parametrization shoud account for the biological constraints:

1. $\alpha_n(v), \alpha_m(v)$ and $\beta_h(v)$ are increasing functions of the voltage,
2. $\beta_n(v), \beta_m(v)$ and $\alpha_h(v)$ are decreasing,.
3. all rates must be positive.

To obtain train data we can integrate (1), (3) with parameters calculated from the real experiments. To learn $\alpha_s(v)$, $\beta_s(v)$ we parameterize them as neural nets and use NeuralODE with L_1 loss function.

Code